

Yash Ravindra Charde

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Summary

Software Engineer with 3+ years of experience designing and building scalable web applications and cloud-native backend systems, and experience with AI, NLP/LLMs, and data structures & algorithms. Skilled in system design and analytical thinking, with a track record of translating complex technical concepts into production-ready solutions.

Education

Master of Science in Computer Science, Arizona State University, Tempe

Aug 2025 – May 2027

• **Coursework:** Cloud Computing, Data Processing at Scale, Data Mining, Statistical Machine Learning, Knowledge Representation, Data Visualization, Natural Language Processing

GPA: 4.00/4.00

Research Publications

- Co-authored "Invariant Reasoning Directions in Latent Trajectories of Language Models" under Dr. Yanjie Fu, improving LLM answer consistency under paraphrase by ~10% and reducing latent trajectory variance by up to 50% across 6 reasoning benchmarks; published (A. V. Malarkkan, M. R. Choudhury, U. Byahut, Y. R. Charde, V. Gupta, Y. Fu et al.).

Experience

Graduate Student Grader, School of Computing and Augmented Intelligence – ASU, Tempe

Aug 2026 – Present

- Evaluate Knowledge Representation & Reasoning graduate coursework for 173 students, spanning classical/first-order logic, answer set programming, ontologies, Semantic Web, knowledge graphs, and the integration of logic with probability, providing feedback aligned with course learning outcomes.

Teaching Fellow / Grader, Computer Information and Cloud Systems – ASU Learning Enterprise

May 2026 – Present

- Serve as Subject Matter Expert and Teaching Fellow for Introduction to Cloud Computing, supporting ~100 scholars and collaborating with NEON, faculty Co-Teachers, and the Course Success Manager to design recitation learning activities.
- Grade assignments and support ~200 learners through Inscribe community discussion boards on cloud computing, information systems, and data analytics concepts, maintaining consistent grading quality.

Software Engineer, ADP Private Limited – Hyderabad, India

Senior Member Technical

Apr 2024 – Jun 2025

- Built a consolidated invoices page and an end-to-end Year-End Readiness Tracker (React.js, Node.js, ASP.NET Web API), delivering responsive, cross-browser dashboards for client-facing billing and compliance workflows.
- Optimized year-end billing and production workflows, cutting billing processing time by 3 hours and completing W-2 production within 6 hours; migrated 48,000+ clients to Local Delivery, generating \$180K in annual savings.

Member Technical

Jul 2021 – Mar 2024

- Developed an event-driven integration platform distributing RUN business events to downstream security consumers (REST, IBM MQ, Kafka), maintaining core processing components for at-least-once delivery across distributed messaging workflows.
- Implemented fraud-alert notifications and led production support and root-cause analysis for security-sensitive event pipelines; modernized the shipping details page, streamlining workflows and improving the user experience.

Projects

Federated Learning System on AWS Cloud and Edge

- Architected an end-to-end Federated Learning pipeline on AWS using 10 distributed EC2 workers and an event-driven Lambda aggregator (triggered via S3) to collaboratively train a LeNet-5 CNN on MNIST, achieving 97.69% accuracy with consistent round-over-round convergence while eliminating polling overhead and preserving data locality.
- Extended the system to edge computing with AWS IoT Greengrass by deploying custom MQTT-triggered components across devices, scoped permissions for secure model uploads and sustained 97.39% accuracy in a Greengrass-based deployment.

Scalable Cloud-Native Face Recognition Pipeline on AWS

- Designed a multi-tier elastic face recognition application on AWS with a stateless EC2 web tier, SQS-decoupled PyTorch inference workers, and S3 + DynamoDB for image storage and result persistence, sustaining sub-second average latency across 100 concurrent requests while cleanly scaling queues from 0 to peak load and back.
- Implemented a custom autoscaling controller in Python that dynamically scales the application tier from 0 to 15 EC2 instances based on SQS queue depth, with DynamoDB result caching delivering ~3x warm-cache speedup.

Technologies

Languages: C/C++, C#, Java, JavaScript (ES6+), TypeScript, Python, PL/SQL, SQL (MySQL, PostgreSQL), HTML, CSS

Frameworks & Libraries: React.js, Angular (v11-v12, TypeScript-based), Node.js, Express.js, ASP.NET Core, Microservices, REST APIs, MVC, Tailwind CSS, Cypress, D3.js, Pandas, NumPy, Matplotlib, OpenCV, Scikit-learn, TensorFlow, PyTorch

Tools & Platforms: AWS (EC2, S3, Lambda, DynamoDB, API Gateway), Docker, Kubernetes (basic), Jenkins, CI/CD pipelines, Git, GitHub, Bitbucket, Agile, Jira, Postman, Splunk, Figma, Linux/Ubuntu